Scaling Book — Section 5 Exercises

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Q1

$$\begin{aligned}
\text{Attn params} &= \underbrace{DNH + 2DKH + NHD}_{\text{attn projs}} + \underbrace{D}_{\text{layernorm}} \\
&= DNH + 2DNH + DNH + D = 4DNH + D. \\
\text{MLP params} &= 2DF + FD = 3DF.
\end{aligned}$$

$$\begin{aligned}
\text{Total} &= (4DNH + D + 3DF) \cdot L + 2DV \\
&= (4 \cdot 5120 \cdot 40 \cdot 128 + 5120 + 3 \cdot 5120 \cdot 13824) \cdot 40 + 2 \cdot 5120 \cdot 32000 \\
&\approx 13 \times 10^9 = 13\text{B params.}
\end{aligned}$$

Q2

BS = 16M tokens, using Adam, store params in bf16, opt state in fp32, checkpoint act 3 times/layer.

$$\begin{aligned}
\text{Param mem} &= 2 \cdot 13 \times 10^9 \text{ bytes} \\
\text{Opt mem} &= \underbrace{4 \cdot 13 \times 10^9 + 4 \cdot 13 \times 10^9}_{\text{1st + 2nd moment}} \\
\text{Act ckpt mem} &= 2L \cdot \Big(\underbrace{2BF}_{\substack{\text{gating} \\ \text{up projs}}} + \underbrace{BD}_{\text{down proj}} \Big) = 2LB(2F + D) \\
&= 2 \cdot 40 \cdot 16 \cdot 10^6 \cdot (2 \cdot 13824 + 5120).
\end{aligned}$$

$$\text{Total} = \text{Param} + \text{Opt} + \text{Act} \approx 4.2 \times 10^{13} \text{ bytes} = 42\text{TB}.$$

Q3

Train with:

- $T = 32\text{K}$
- $\text{BS} = 3\text{M}$

- on TPU v5p $16 \times 16 \times 16$ slice (has wraparound conns)
- bf16 weights, fp32 optimizer

(i)

From prev q, assuming Adam optimizer, the model weights + opt state is 130 GB > 96 GB which is the v5p's memory capacity.

(ii)

With ZeRO-3 type FSDP, we have no duplicated data.

$$\begin{aligned}\text{Bytes}_{\text{Total}} &= (10 \cdot 13 \cdot 10^9) + 2LB(2F + D) \\ &= (10 \cdot 13 \cdot 10^9) + 2 \cdot 40 \cdot (3 \cdot 10^6) \cdot (2 \cdot 13824 + 5120) \\ &\approx 7.99 \cdot 10^{12} \text{ bytes} \approx 8\text{TB}.\end{aligned}$$

Total v5p $16 \times 16 \times 16$ slice has $16^3 \cdot 96$ GB = 393 TB, so we are in the hbm limit. Each device will have $(8\text{TB})/16^3 \approx 1.96$ GB used. $\quad (X = 16^3)$

$$T_{\text{comm}} = \frac{2 \cdot 2DF}{3 \cdot W_{\text{ici}}}, \qquad T_{\text{math}} = \frac{2 \cdot 2BDF}{X \cdot C}.$$

$$\Rightarrow \frac{B}{X} = \frac{3 \cdot 10^6}{16^3} = 732 \;<\; \frac{4.59 \cdot 10^{14}}{3 \cdot 1.8 \cdot 10^{11}} = 850.$$

Using pure FSDP, while it is possible, we will be comm bounded so it's better to try FSDP+TP.

(iii)

a) To be compute bound:

$$\frac{B}{N} \;>\; \frac{X^2}{\underbrace{M_x M_y}_{\text{2 for 3D mesh}} F} = \frac{(2550^2)}{2 \cdot 13824} \approx 235.$$

$$\frac{B}{N} = 850 > 235, \text{ so we can use FSDP + TP.}$$

b) Choosing M_x, M_y doesn't matter $\Rightarrow$ just changes X_{opt}.

Let $M_x = 2$, $M_y = 1$.

$$X_{\text{opt}} = \sqrt{\frac{B}{F} \cdot \frac{M_x}{M_y} \cdot N} = \sqrt{\frac{3 \cdot 10^6}{13824} \cdot 2 \cdot 16^3} \approx 1333.$$

$$\Rightarrow \text{mesh} = \{X{:}1024,\; Y{:}4\} \qquad \underbrace{}_{\text{need int + want wraparounds}}$$

FSDP+TP sharding:

$$\mathrm{In}[B_x, D_y] \cdot_D W_{\mathrm{in}}[D_x, F_y] \cdot_F W_{\mathrm{out}}[F_y, D_x] \to \mathrm{Out}[B_x, D_y].$$

Per device mem = 1.96 GB   (same as pt b since there's no data duplication).

c) No redundant comp w/ FSDP+TP:

$$\mathrm{FLOPs}_{\mathrm{Total}} = 6 \cdot 13\mathrm{B} \cdot 3\mathrm{M} = 6 \cdot 13 \cdot 10^9 \cdot 3 \cdot 10^6.$$

$$T_{\mathrm{comp}} = \frac{(6 \cdot 13 \cdot 10^9 \cdot 3 \cdot 10^6)}{16^3 \cdot 4.59 \cdot 10^{14} \cdot 0.4} = 0.3.$$